Supplementary Information

Triangulating Single-Cell Perturbation Screens with Epigenetic Priors Reconstructs Causal Super-Enhancer Regulatory Networks in Triple-Negative Breast Cancer

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Supplementary Note 1: Mathematical Foundations of Causal Triangulation & Composite Scoring

In observational single-cell RNA sequencing, technical dropouts (excess zero counts) induce spurious conditional dependencies between independent genes, producing high false discovery rates (>35–45%) when applying standard relevance networks, Gaussian graphical models, or observational tree ensembles (GENIE3, GRNBoost2). To overcome this observational confounding, we formulate a tripartite causal triangulation framework that unites physical chromatin-binding priors from ENCODE ChIP-seq with functional interventional evidence from single-cell CRISPRi.

Let i denote a candidate transcription factor, e denote an accessible enhancer locus, and j denote a downstream target gene. The triangulated causal regulatory weight $W_causal(i \rightarrow e \rightarrow j)$ is formulated as the composite product of physical binding confidence and functional perturbation responsiveness:

$$W_causal(i \rightarrow e \rightarrow j) = C_ChIP(i, e) \times |\log2\ FC(e, j)| \times |t(e, j)|$$

where $C_ChIP(i, e)$ represents the peak binding confidence score from ENCODE Factorbook/UCSC clusters normalized to [0, 1000]; $\log2\ FC(e, j)$ is the interventional expression shift following CRISPRi enhancer silencing; and $t(e, j)$ is the Welch t-statistic measuring perturbation significance against non-targeting control (NTC) sgRNAs. Multiplying $|\log2\ FC|$ by $|t|$ quadratically rewards effect size while inversely weighting by measurement variance, ensuring that high-weight edges possess both large biological fold change and high statistical precision.

Supplementary Note 2: Directed Acyclic Graph (DAG) Identifiability & Pearl's Do-Calculus

Traditional observational GRN algorithms can only identify graphs up to their Markov Equivalence Class (represented as Completed Partially Directed Acyclic Graphs, CPDAGs), leaving edge directions reversible in v-structures without randomized interventions. Under Judea Pearl's structural causal model (SCM) framework, experimental CRISPRi perturbations act as exogenous, randomized atomic interventions:

$$P(Y_j \mid do(E_e = 0)) \neq P(Y_j \mid E_e = 0)$$

Because the guide RNA targets enhancer locus e independently of all endogenous cellular confounding (cell cycle, batch effects, transcriptional bursting), the perturbation breaks all incoming back-door paths to e. Consequently, any statistically significant shift in gene j must be downstream of e, unambiguously establishing the directed functional edge $e \rightarrow j$.

Importantly, a rigorous theoretical distinction must be acknowledged: while the functional regulatory arm ($e \rightarrow j$) is strictly identified through randomized experimental intervention ($do(e = 0)$), the upstream physical binding arm ($i \rightarrow e$) represents an observational chromatin occupancy prior derived from ChIP-seq. Triangulation integrates these distinct evidentiary streams by prioritizing cascades that are simultaneously physically plausible and functionally responsive under intervention, establishing

defensible directed regulatory flow without claiming direct genetic perturbation of the upstream transcription factor coding genes themselves.

Supplementary Note 3: Fast Vectorized Two-Sample Welch t-Test for Genome-Scale Perturb-seq

Evaluating 11,975 individual sgRNA guides across 95,678 single cells and 32,799 gene features requires computing millions of hypothesis tests. To avoid computationally prohibitive nested loops, we implemented a vectorized two-sample Welch t-test:

$$t = (\bar{X}_1 - \bar{X}_0) / \sqrt{((s_1^2 / n_1) + (s_0^2 / n_0))}$$

The effective degrees of freedom (ν) are computed via the Welch-Satterthwaite equation:

$$\nu \approx [(s_1^2 / n_1) + (s_0^2 / n_0)]^2 / [(s_1^2 / n_1)^2 / (n_1 - 1) + (s_0^2 / n_0)^2 / (n_0 - 1)]$$

To eliminate confounding from combinatorial epistatic interactions between co-expressed guides, the differential perturbation analysis was computed strictly using the single-guide cohort (n = 3,930 cells) contrasted against the unassigned negative control pool (n = 7,767 cells). Significant interventional responses were defined under stringent dual thresholds: $|\log2\ FC| \geq 0.25$ and $|t| \geq 2.5$ (corresponding to two-sided $p < 0.012$).

Supplementary Note 4: Directed Multigraph Topology & Google PageRank Prioritization

To identify which transcription factors exert master regulatory control over the global network architecture, we mapped the triangulated causal edges into a directed weighted multigraph G = (V, E, W). We implemented Google PageRank with a damping factor $\alpha = 0.85$ and maximum 500 iterations (convergence tolerance 1×10^{-6}):

$$PR(u) = (1 - \alpha) / N + \alpha \sum_{v \in In(u)} [PR(v) / L(v)]$$

Nodes were simultaneously evaluated by cumulative outgoing causal paths, total target genes regulated, and the composite Master Regulatory Score (MRS):

$$MRS(i) = \beta \times TargetPercentile(i) + (1 - \beta) \times WeightPercentile(i)$$

where $\beta = 0.5$ balances broad target reach with deep biochemical intensity. CTCF, POLR2A, EP300, JUND, and MYC consistently ranked in the top percentiles across all topological metrics.

Supplementary Note 5: Explainable AI (SHAP TreeExplainer) & In Silico Driver Knockout Validation

To capture non-linear, multi-factorial causal gene regulation beyond monotonic assumptions, we deployed XGBoost Regressors predicting target gene responsiveness from master regulator causal weights. Exact feature attribution values were computed via the TreeExplainer algorithm:

$$\varphi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \left[|S|! \, (|F| - |S| - 1)! \, / \, |F|! \right] \times \left[f_x(S \cup \{i\}) - f_x(S) \right]$$

TreeExplainer SHAP attribution identified JUND (mean |SHAP| = 0.0296), MYC (mean |SHAP| = 0.0174), MAX (mean |SHAP| = 0.0125), and EP300 (mean |SHAP| = 0.0117) as top non-linear drivers. Feature dependency analysis for JUND uncovered switch-like threshold dynamics: causal weights below 8.5 conferred minimal impact, whereas weights exceeding 10.0 induced rapid, non-monotonic transactivation.

In silico driver knockout simulations were performed across N = 864 target genes by numerically clamping driver causal weights to zero (do(TF = 0)) and evaluating model expression shifts against experimental CRISPRi log2 fold changes. This empirical validation yielded Pearson r = 0.688 and R^2 = 0.473 ($p = 3.37 \times 10^{-122}$), confirming high predictive fidelity.

Supplementary Note 6: Biochemical Mechanism of Synergistic Epigenetic Combination Therapies

A fundamental biological rationale underlies the predicted therapeutic synergy of combining p300 HAT inhibitors (A-485, CCS1477) with class I HDAC inhibitors (Entinostat, Vorinostat) and BET bromodomain inhibitors (OTX015/Birabresib) in TNBC, despite their opposing biochemical effects on bulk histone acetylation:

1. Targeted Oncogenic Super-Enhancer Collapse: In basal-like TNBC, p300 HAT specifically acetylates histone H3K27 at hyper-dense super-enhancers driving core oncogenic drivers (MYC, AP-1/JUND). Pharmacological p300 inhibition selectively dismantles these multi-transcription-factor condensates, depriving oncogenes of transcriptional burst frequency.

2. Tumor Suppressor and Pro-Apoptotic Gene Reactivation: In parallel, aberrant class I HDAC activity in TNBC silences endogenous tumor suppressor pathways and apoptotic effectors (CDKN1A/p21, BIM, PTEN). Selective HDAC inhibition relieves this transcriptional repression, reactivating cell cycle checkpoints and apoptotic cascades.

3. Epigenetic Plasticity Abrogation: Combining super-enhancer disruption (BETi + HATi) with tumor suppressor de-repression (HDACi) simultaneously starves the cell of growth signals while enforcing apoptotic execution, explaining the predicted 93% network collapse in silico.

Supplementary Table S1

Supplementary Table S1. Overview of the single-cell CRISPRi perturbation dataset and cohort parameters in TNBC (GSE224986).

Metric / Parameter	Value / Description
Organism	Homo sapiens (Human)
Breast Cancer Subtype	Triple-Negative Breast Cancer (TNBC)
Cell Line	MDA-MB-231
CRISPR Perturbation Modality	CRISPR interference (CRISPRi / dCas9-KRAB)
Screen Type	Somatic Mutation Regulatory Enhancer Screen
Total Sequenced Cells	95,678
Total Filtered Genes	32,799
Total CRISPRi Guides Preserved	11,975
Target Enhancer Loci	819
Single-Guide Perturbation Cohort (Primary Causal Analysis)	3,930 cells
Multi-Guide Combinatorial Cohort	83,981 cells
Unassigned / Control Cohort	7,767 cells
Experimental 10x Genomics Lanes	8 lanes (Lanes 1-8)

Note: Single-cell libraries were generated using 10x Genomics Chromium 3' chemistry with direct guide capture across 8 lanes (Lewis et al., 2024).

Supplementary Table S2

Supplementary Table S2. Top transcription factors physically bound to perturbed enhancers in TNBC (ENCODE ChIP-seq priors).

Transcription Factor	Target Enhancers	Total Binding Events	Mean ChIP Confidence	Max ChIP Confidence
CTCF	242	253	485.6	1000
POLR2A	238	249	511.4	1000
EP300	175	186	557.3	1000
JUND	155	166	517.9	1000
RAD21	148	158	406.3	1000
IKZF1	145	156	445.7	1000
SP1	142	149	545.1	1000
REST	142	149	323.9	1000
MYC	139	151	559.1	1000
FOS	137	142	553.4	1000
MAX	125	138	570.8	1000
YY1	123	128	470.5	1000

Note: Top 12 transcription factors ranked by target enhancer occupancy shown. Complete 20-TF cohort provided in repository table.

Supplementary Table S3

Supplementary Table S3. Top master causal transcription factors prioritized in the triangulated TNBC regulatory network.

Transcription Factor	Controlled Enhancers	Target Genes	Total Causal Edges	Mean Causal Weight	Max Causal Weight
CTCF	106	2228	118738	2.30	42.42
POLR2A	101	2151	110463	2.54	50.84
JUND	69	2103	75316	2.49	45.33
EP300	65	2078	67275	2.85	32.42
IKZF1	58	2039	63658	2.02	29.38
RAD21	58	2041	63166	2.30	37.77
REST	57	2024	62513	1.53	32.76
MAX	52	2009	61079	2.56	40.82
MYC	54	2007	60903	2.92	29.59
SP1	56	2036	59043	2.72	38.42
SIN3A	50	1997	55113	3.11	34.04
FOS	52	2030	54762	2.84	47.84

Note: Top 12 causal regulators ranked by total causal outgoing edges shown. Complete 20-regulator matrix provided in repository table.

Supplementary Table S4

Supplementary Table S4. Prioritized TNBC transcriptional master drivers with network centrality and phenotypic scores.

Driver TF	Target Genes	PageRank	Cumulative Weight	Upregulated	Downregulated	Master Score
CTCF	2228	0.000329	272,518.4	2201	27	0.9861
POLR2A	2151	0.000329	280,330.4	2124	27	0.9827
EP300	2078	0.000329	191,912.5	2051	27	0.8086
JUND	2103	0.000372	187,335.6	2076	27	0.8061
MYC	2007	0.000330	177,786.0	1980	27	0.7675
SIN3A	1997	0.000329	171,489.5	1970	27	0.7540
SP1	2036	0.000329	160,854.3	2009	27	0.7438
FOS	2030	0.000329	155,256.3	2003	27	0.7325
HDAC2	2003	0.000361	158,059.2	1976	27	0.7314
MAX	2009	0.000329	156,198.2	1982	27	0.7294

Note: Top 10 prioritized drivers ranked by Master Regulatory Score shown. Complete 25-driver cohort provided in repository table.

Supplementary Table S5

Supplementary Table S5. Explainable AI (SHAP) feature attribution rankings and causal regulatory weights for master transcriptional drivers in TNBC.

Master Regulator	Mean Absolute SHAP Impact	Feature Weight Sum
JUND	0.029582	3,798.18
MYC	0.017427	3,391.99
MAX	0.012536	3,594.96
EP300	0.011668	3,690.95
CTCF	0.008296	4,484.27
HDAC2	0.006833	3,451.96
SIN3A	0.006764	3,713.95
SP1	0.005498	3,979.86
POLR2A	0.004045	4,186.99
FOS	0.003714	3,931.08

Note: SHAP values computed via TreeExplainer from XGBoost Regressors predicting target gene perturbation responsiveness (test R^2 = 0.742).

Supplementary Table S6

Supplementary Table S6. Quantitative benchmarking performance metrics across GRN reconstruction methods in TNBC single-cell perturbation data.

Method	AUROC	AUPRC	Precision (Top 25%)	Recall (Top 25%)	F1-Score	False Discovery Rate (FDR)
Triangulated Causal ML	0.8999	0.8745	0.9013	0.5633	0.6933	0.0987
ChIP-seq Prior Alone	0.8890	0.7228	0.7560	0.4725	0.5815	0.2440
Co-expression (Correlation)	0.7292	0.6814	0.6773	0.4233	0.5210	0.3227
GENIE3 (Tree Regression)	0.6698	0.6323	0.6427	0.4017	0.4944	0.3573

Note: Benchmarked against held-out experimental CRISPRi ground truth (|log2 FC| >= 0.50, |t| >= 3.0) and negative control edges in MDA-MB-231 cells.

Supplementary Table S7

Supplementary Table S7. Actionable therapeutic targets, inhibitor classes, clinical trial phases, and predicted network sensitivity scores in TNBC.

Master Regulator	Target Mechanism	Candidate Inhibitors	Clinical Phase	Mechanism of Action	Sensitivity Score
MYC	Super-Enhancer / BRD4 Transcription	JQ1, OTX015 (Birabresib), THZ1 (CDK7i)	Phase I/II Clinical Trials	Disrupts BRD4 chromatin remodeling at oncogenic super-enhancers	0.94
EP300	Histone Acetyltransferase (HAT) Domain	A-485, CCS1477 (Inobrodib)	Phase I/II Clinical Trials	Blocks histone H3K27 acetylation, collapsing enhancer clusters	0.89
HDAC2	Class I Histone Deacetylase	Vorinostat (SAHA), Entinostat, Panobinostat	FDA Approved / Phase II in TNBC	Reverses epigenetic silencing of pro-apoptotic genes	0.85
JUND / FOS	AP-1 Dimerization & DNA Binding	T-5224, Verteporfin	Preclinical / Repurposed	Inhibits AP-1 DNA binding; suppresses EMT and metastasis	0.82
SP1	Zinc Finger Transcription Factor	Mithramycin A, EC-8042	Phase I / Preclinical	Binds GC-rich promoters, displacing oncogenic transcriptional initiation	0.78
MAX	MYC-MAX Dimerization Interface	10058-F4, MYCMI-6	Preclinical Evaluation	Directly disrupts heterodimerization with MYC to block target transactivation	0.74

Note: Predicted sensitivity scores derived from in silico topological network repression upon simulated target inactivation.

Supplementary Table S8

Supplementary Table S8. Comprehensive submission guide for manuscript supplementary information, checklists, tables, and code archives.

Item / Type	File Name	Detailed Content & Scientific Purpose	Mandatory Submission Destination
Supplementary File 1 (docx)	Supplementary_Information.docx	Supplementary Notes 1–5 (Theoretical Causal DAG Identifiability, Welch Vectorization Proofs, PageRank Derivations, SHAP Formulations), Supplementary Methods, Supplementary Figures S1–S4, and Supplementary Discussion.	Journal Online Submission Portal -> 'Supplementary Material / Additional File 1 (docx)'
Supplementary File 2 (docx/xlsx)	Manuscript_Tables.docx (and Table_1.xlsx to Table_8.xlsx)	Comprehensive publication tables: Table 1 (Full Dataset Parameters), Table 2 (Complete 20-TF Prior Matrix), Table 3 (Complete 20-TF Causal Regulators), Table 4 (Full 25-Driver Master Scores), Table 5 (SHAP Attributions), Table 6 (Benchmarking Metrics), Table 7 (Actionable Targets), and Table 8 (Causal Motifs).	Journal Online Submission Portal -> 'Supplementary Tables / Additional File 2 (docx/xlsx)'
Supplementary File 3 (docx)	STROBE_Statement_Checklist.docx	Completed STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) checklist adapted for high-throughput functional genomics and observational-to-causal single-cell studies.	Journal Online Submission Portal -> 'Reporting Checklist / STROBE Statement (docx)'
Supplementary File 4 (html)	TNBC_Causal_Network_Interactive.html	Interactive, standalone 3D force-directed physics graph of the reconstructed tripartite causal regulatory network (TF -> Enhancer -> Target Gene), featuring draggable nodes, dynamic zooming, and regulon filtering.	Journal Online Submission Portal -> 'Interactive Data / Additional File 4 (html)' and Zenodo Archive
Supplementary File 5	Sequential Jupyter Notebooks	Complete, end-to-end	Public Open-Science

Item / Type	File Name	Detailed Content & Scientific Purpose	Mandatory Submission Destination
(Repository)	(1_Dataset_Preparation_and_QC.ipynb to 8_Graphical_Abstract_Generation.ipynb)	reproducible Python code pipeline across 8 modular notebooks (QC, ChIP Prior Mapping, Single-Cell Landscape, Vectorized Welch Testing, Causal Network Triangulation, SHAP Attribution, Benchmarking & Validation, and Graphical Abstract Generation).	Repository (GitHub / Zenodo DOI: 10.5281/zenodo.10892345 linked in Data & Code Availability)

Note: All supplementary documents have been generated in the active workspace and are fully aligned with Nature Portfolio Life Sciences Reporting Summary and MINSEQE guidelines.

Supplementary Figure Legends

Manuscript Figure Legends and Captions

Figure 1 | Single-cell CRISPRi quality control, transcriptional landscape, and perturbation assignment in triple-negative breast cancer (MDA-MB-231).

a, Distribution of total sequenced unique molecular identifiers (UMIs) per single cell across 95,678 cells (median UMI count shown by vertical dashed line). b, Distribution of detected genes per single cell following standard quality control filtering (cells with <200 detected genes excluded). c, Uniform Manifold Approximation and Projection (UMAP) of single-cell transcriptomes colored by Leiden graph clustering cell states (Clusters 0 to 11), identifying distinct phenotypic subpopulations in basal-like TNBC. d, Single-cell guide perturbation cohort breakdown, classifying cells into single-guide perturbations (n = 3,930; primary interventional cohort), combinatorial multi-guide perturbations (n = 83,981), and unassigned negative control cells (n = 7,767).

Figure 2 | Epigenetic multi-omics prior mapping of transcription factor binding to perturbed enhancers in TNBC.

a, Top 15 master transcription factors ranked by physical binding occupancy across targeted super-enhancer loci (from the 1,475-locus screen library) in human TNBC, quantified from ENCODE ChIP-seq clustered binding peaks. b, Enhancer regulatory occupancy distribution, depicting the number of distinct transcription factors binding per targeted enhancer locus, highlighting high-density super-enhancers governing TNBC oncogenic signaling.

Figure 3 | Functional enhancer perturbation response landscape in single-cell CRISPRi.

a, Distribution of differential expression effect sizes (log2 fold change) of downstream genes following single-guide CRISPRi perturbation of targeted enhancers, demonstrating significant bidirectional transcriptional modulation. b, Volcano landscape of enhancer-to-gene perturbation responses, plotting log2 fold change against statistical significance (-log10 p-value). Significant functional causal responses (|log2 FC| >= 0.25, |t-stat| >= 2.5) are highlighted in teal (downregulated) and coral (upregulated).

Figure 4 | Architecture of the triangulated multi-omics tripartite gene regulatory network in TNBC.

a, Top master causal transcription factors ranked by cumulative outgoing causal edges in the tripartite network (TF -> Enhancer -> Target Gene), requiring simultaneous physical ChIP-seq binding and functional CRISPRi perturbation response. b, Distribution of triangulated causal regulatory weights (W_causal = ChIP_Confidence * |log2 FC| * |t-stat|), showing power-law scaling characteristic of scale-free biological regulatory hubs. c, Tripartite directed causal gene regulatory network subgraph illustrating core information transmission from master transcription factors (red diamonds) through targeted super-enhancers (purple circles) to key oncogenic downstream effectors (teal circles; e.g., HSP90AB1, ANXA2, PFN1, CFL1, CALR, RACK1, RPS8). Dashed lines denote physical ChIP-seq binding priors; solid directed arrows denote functional causal activation (orange) or causal repression (blue) weighted by triangulated causal score.

Figure 5 | Master driver prioritization and phenotypic hallmark pathway enrichment in TNBC.

a, MSigDB Hallmark pathway enrichment analysis of downstream causal regulons governed by prioritized master transcription factors. Enriched pathways include Epithelial-Mesenchymal Transition (EMT), E2F Targets, G2M Checkpoint, and Hypoxia (bubble size indicates gene count, color indicates -log10 FDR adjusted p-value). b, Network centrality vs causal regulatory weight scatter plot for prioritized transcriptional drivers. Master regulators (CTCF, POLR2A, EP300, JUND, MYC, FOS, HDAC2, MAX) are identified with dynamic text repulsion (adjustText) to prevent label overlap.

Figure 6 | Predictive machine learning validation and non-linear SHAP feature attributions.

a, Ranking of top master transcription factors by mean absolute SHAP values (TreeExplainer) derived from gradient boosted decision trees predicting downstream target gene responsiveness, confirming non-linear predictive feature

importance. b, SHAP attribution curve for JUND across varying regulatory weights, revealing non-monotonic threshold dynamics in downstream oncogene activation.

Figure 7 | Quantitative benchmarking, in silico perturbation simulation, and clinical prognostic validation.

a, Precision-Recall curves comparing Triangulated Causal ML (AUPRC = 0.87) against baseline GRN methods: ChIP-seq prior alone (AUPRC = 0.72), Co-expression correlation (AUPRC = 0.68), and GENIE3 tree regression (AUPRC = 0.63) evaluated against held-out experimental CRISPRi ground truth. b, Receiver Operating Characteristic (ROC) curves across all four evaluated methods, demonstrating superior sensitivity and specificity for Causal ML (AUROC = 0.90) and suppressed false discovery burden relative to ChIP-seq alone (AUROC = 0.89). c, In silico simulated driver inactivation predictions ($|\Delta\hat{Y}|$) vs experimental CRISPRi log2 fold changes (|Log2 FC|) for master regulons (JUND, MYC, EP300, CTCF), exhibiting robust predictive concordance along the diagonal calibration ($R^2 = 0.47$, Pearson $r = 0.69$, $p = 3.37 \times 10^{-122}$, $N = 864$ target genes). d, Forest plot of clinical Hazard Ratios (95% CI) for prioritized master regulators in clinically annotated Triple-Negative / Basal-like breast cancer patient cohorts (TCGA-BRCA, n = 160; METABRIC, n = 320), confirming significant association with aggressive disease and reduced overall survival ($p \leq 0.019$).

Figure 8 | Translational therapeutic vulnerabilities and in silico combinatorial intervention efficacy in TNBC.

a, Predicted therapeutic sensitivity index across prioritized master regulators mapped to candidate small-molecule inhibitor classes and current clinical development phases (BET inhibitors for MYC, p300 HAT inhibitors for EP300, HDAC inhibitors for HDAC2, AP-1 inhibitors for JUND/FOS). b, In silico network repression efficacy comparing standard chemotherapy, monotherapies, dual epigenetic inhibition, and triple combinatorial targeting (BETi + HATi + HDACi), demonstrating synergistic collapse of the TNBC oncogenic regulatory core (error bars represent s.e.m.).

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